3D MODEL · SPEC CONVERSION

STEP → E3D Equipment

Heuristically recognizes NX STEP B-rep shapes and converts them to E3D primitives (CYLI / BOX / cylinder). Handles nozzles / flanges / bolts.

Key Purpose

Why this program

1. **Turn a vendor STEP model into an E3D Equipment macro**

   – Reads the NX B-rep solid assembly and writes a ready-to-run .mac that builds EQUI / SUBE / primitives

   → **Effect:** Supplier 3D equipment is re-modelled in minutes instead of hours of manual primitive drawing, with no transcription error

2. **Recognize shapes automatically as native primitives**

   – Analyzes each solid's surfaces and maps them to CYLI / BOX / CONE / DISH / CTORUS — flanges, nozzles, bolts, elbows, structural sections

   → **Effect:** The converted model uses real E3D primitives that can be edited and clash-checked, not a dumb mesh

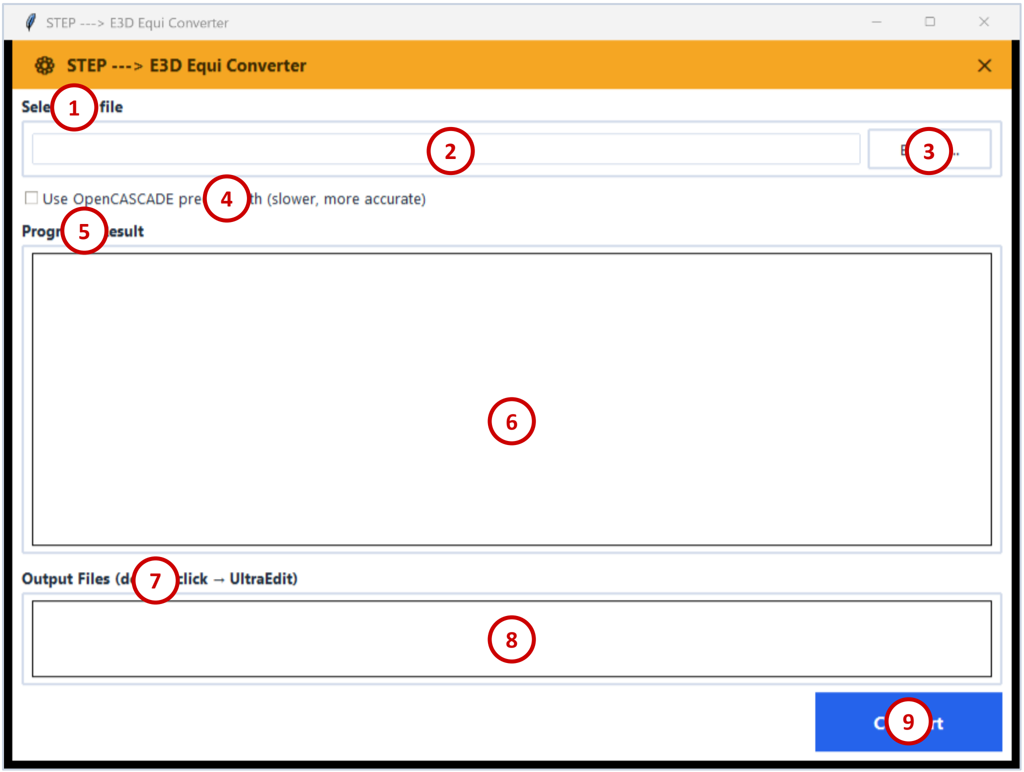
3. **Keep true world position per part**

   – Walks the STEP assembly tree and preserves each occurrence's absolute placement

   → **Effect:** Every part lands where it belongs, so the equipment drops straight into the plant model without re-fitting

Required inputs : A STEP (.stp) B-rep solid assembly file

Main Window – Item Guide



1. Select STP file
2. STP file path
3. Browse for STP file
4. Use OpenCASCADE precise path (slower, more accurate)
5. Progress / Result
6. Progress / Result log
7. Output Files (double-click to open)
8. Output files list
9. Convert

Overview

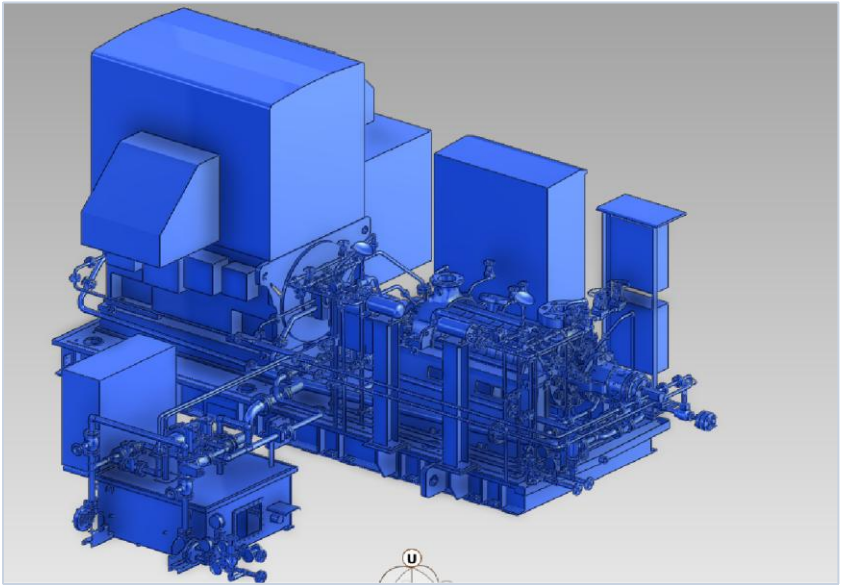
What it does

1. **Heuristically recognizes NX STEP B-rep shapes and converts them to E3D primitives (CYLI**
2. **BOX**
3. **Cylinder)**
4. **Handles nozzles**
5. **Flanges**
6. **Bolts**

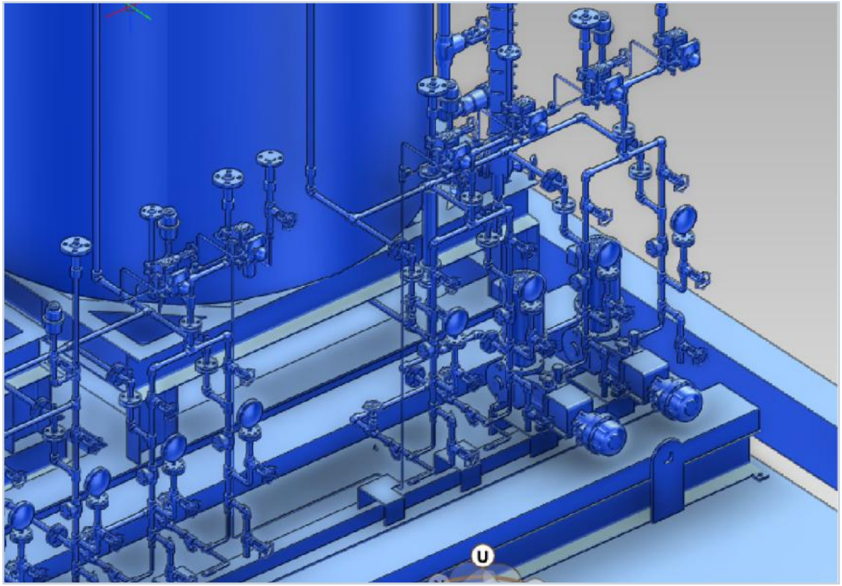
Category: 3D Model · Spec Conversion
Folder: E3D_STP_CONV
Main scripts: stp_to_e3d_equi.py
Scripts / Status: 4 scripts · Experimental
Tags: STEP · B-rep · E3D · primitive

Output Samples

Output Samples (1/4)



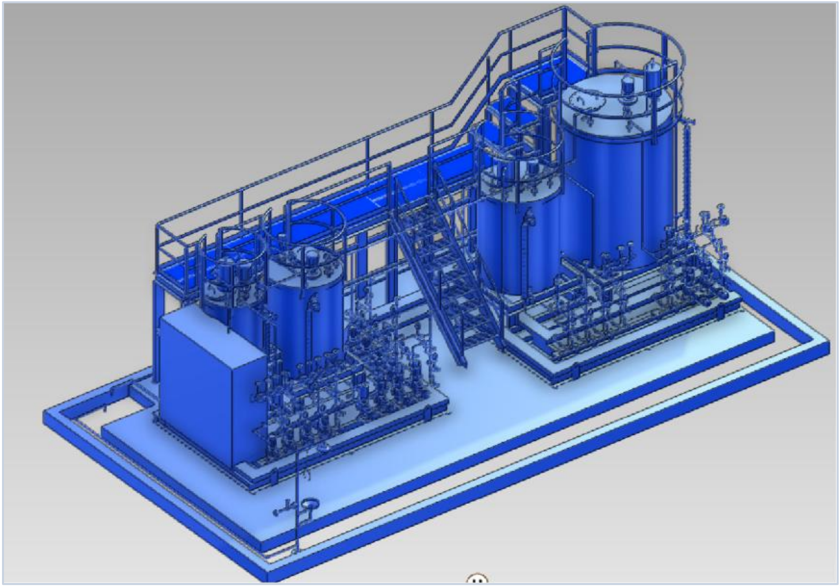
INPUT1



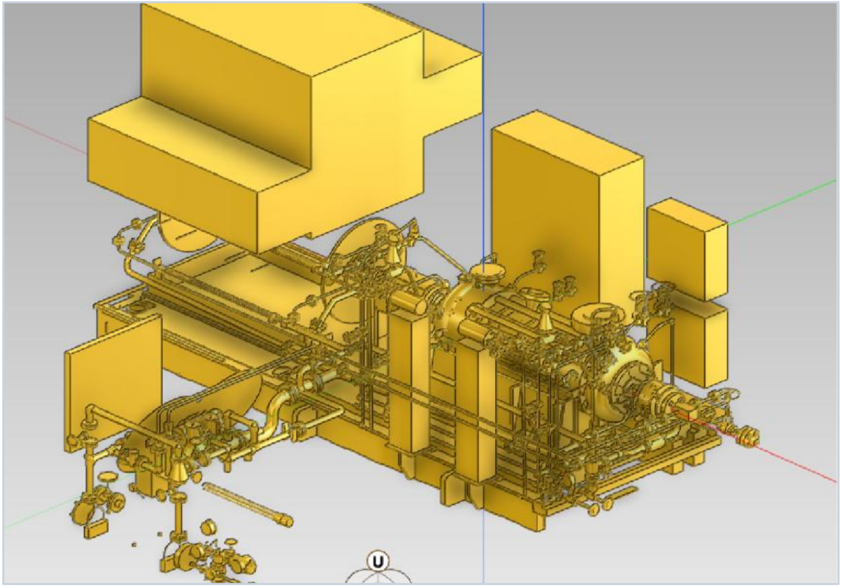
input10-1

Output Samples

Output Samples (2/4)



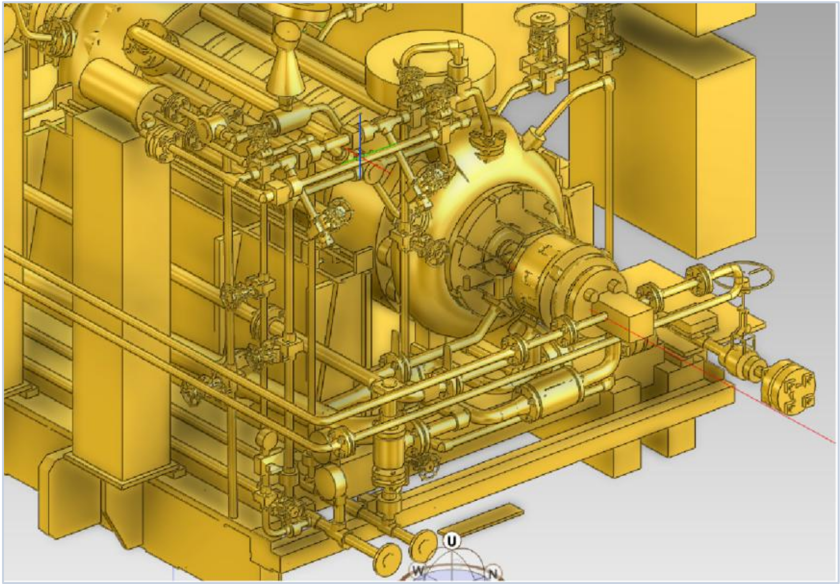
input10



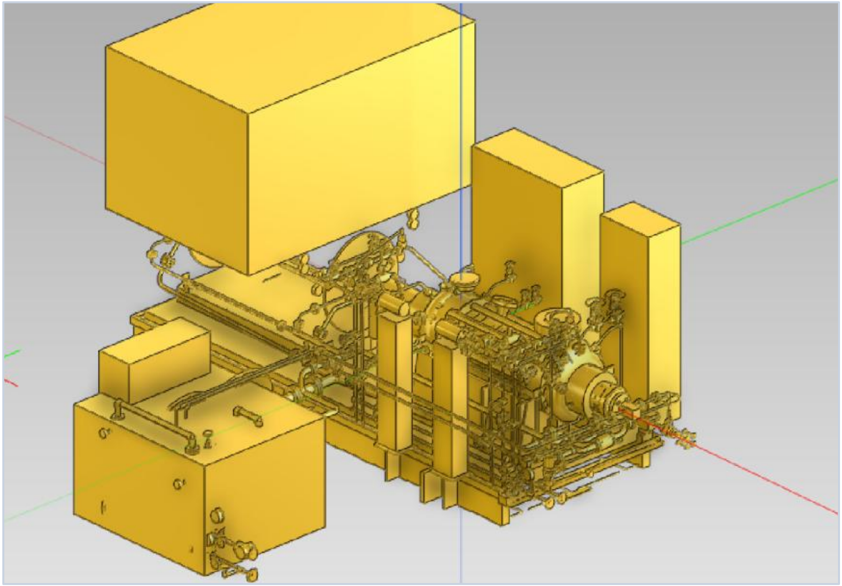
OUTPUT1-1

Output Samples

Output Samples (3/4)



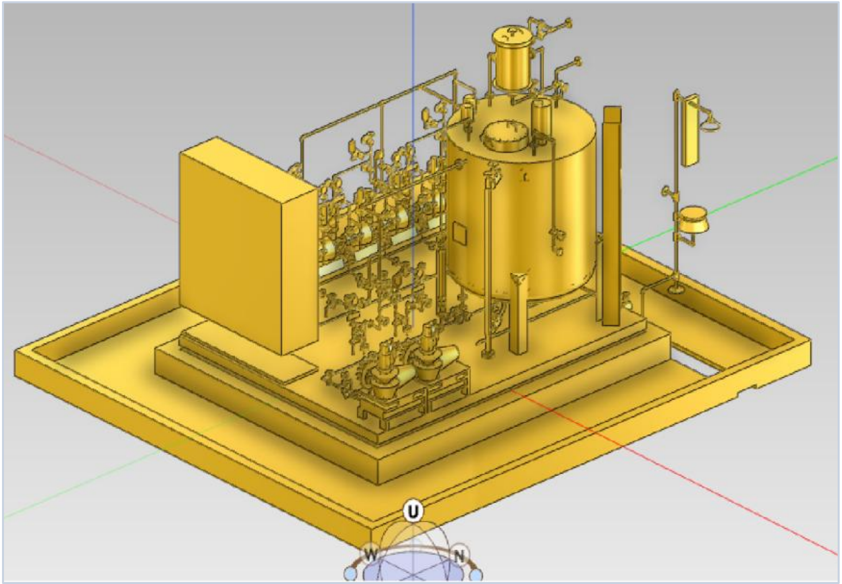
OUTPUT1-2



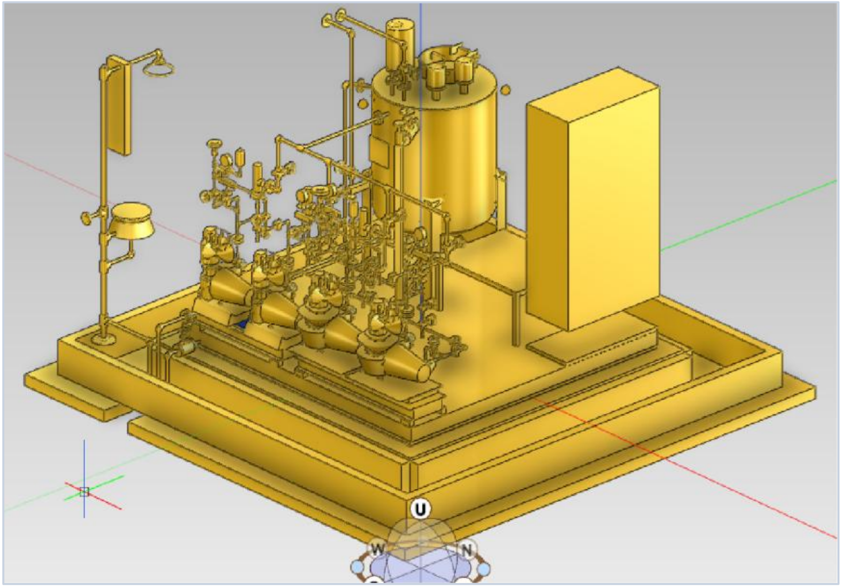
OUTPUT1

Output Samples

Output Samples (4/4)



OUTPUT2



OUTPUT3

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: 3D Model · Spec Conversion.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- Launch from the PC launcher (런처.bat) or run the main script: stp_to_e3d_equi.py.
- Current status: Experimental — behavior/outputs may still change.
- Output samples and the program window above reflect the current build.

Thank You

STEP → E3D Equipment · Plant Engineering Total Service System